Lesson 166: Automating a Simple Task with Macro Recorder

Section 31: Automation of a Few Projects with Macros | Course Video 244 | 15-minute lecture

Learning goal: Record repetitive Excel actions, inspect the recorded VBA, use AI to help make the task reusable, save a macro-enabled workbook, run it safely, and add a command to the Excel Ribbon.

Instructor example: An SEO dataset contains unwanted trailing emojis in keywords and descriptions. The instructor manually cleans a few cells, records the operation, asks an AI assistant to generalize the VBA, and runs the revised macro.

1. What are automation and macros?

Automation is using software to complete repeated tasks with less manual work. A macro is a reusable set of instructions. In desktop Excel, the Macro Recorder generates VBA (Visual Basic for Applications) code from supported user actions.

Concept	Meaning
Macro Recorder	Records supported actions into VBA instructions
VBA Editor	Where you inspect or edit the generated macro
AI assistant	Can help explain, revise, and generalize recorded code
Ribbon customization	Lets you launch a macro using a named command

2. Enable the Developer tab

1. Open File > Options.

2. Choose Customize Ribbon.

3. Under Main Tabs, select Developer.

4. Click OK. Excel now displays the Developer tab.

3. Record the sample task

Practice data: A2 = Learn Excel [emoji], A3 = Python Training [emoji], A4 = Data Analytics [emoji]. The emojis are represented here as descriptions so you can substitute any unwanted trailing emojis in your own test file.

1. Select the worksheet containing your sample SEO data.

2. Choose Developer > Record Macro.

3. Enter the macro name Removing_Emoji. Macro names cannot contain spaces.

4. Choose This Workbook as the storage location and click OK.

5. Edit A2, A3 and A4 manually to remove the unwanted trailing emojis.

6. Choose Developer > Stop Recording.

What you should expect

Macro Recorder captures actions on the particular cells you edited. Replaying that recording may edit just those same cells, rather than automatically scanning a growing dataset. The instructor uses AI to address this limitation.

4. Inspect the recorded VBA

1. Select Developer > Visual Basic, or press Alt + F11.

2. In Project Explorer, expand the workbook and its Modules folder.

3. Open the module containing Removing_Emoji.

4. Read the statements; do not delete unrelated procedures.

Illustrative code only - your recorded code will differ:

```
Sub Removing_Emoji()
    Range("A2").Value = "Learn Excel"
    Range("A3").Value = "Python Training"
    Range("A4").Value = "Data Analytics"
End Sub
```

VBA text	Meaning
Sub Removing_Emoji()	Begins the named macro
Range("A2")	References cell A2
.Value = ...	Replaces the cell value
End Sub	Ends the procedure

5. Use AI to improve the recorded macro

The instructor provides the AI assistant with the problem, the observed emoji pattern, and the recorded VBA. The goal is to process matching text across the active sheet, rather than only a few hard-coded cells.

Reusable prompt

I have an Excel SEO dataset where unwanted emojis appear at the end of keywords or descriptions. I recorded a VBA macro while removing them from a few sample cells. Here is the code: [paste recorded VBA]. Please help generalize it to the active worksheet, regardless of row or column. Preserve legitimate text and punctuation; skip formulas and error cells; handle Unicode emoji safely; and explain how to test it on a copy before running it.

Source limitation: The provided lecture transcript describes the final AI-generated macro but does not reproduce its complete VBA. This guide therefore does not present an unverified universal emoji-removal algorithm.

6. Replace the code and save correctly

1. Review the generated code carefully; test it on a duplicate workbook.

2. Return to the VBA Editor (Alt + F11) and open the relevant module.

3. Replace only the intended macro procedure with the revised code.

4. Choose File > Save As and select Excel Macro-Enabled Workbook (*.xlsm).

5. Save the workbook. Standard .xlsx files do not retain embedded VBA macros.

Extension	VBA support
.xlsx	No
.xlsm	Yes; used in this lesson
.xlsb	Yes
.csv	No

7. Run the macro

1. Make sure the intended SEO worksheet is active.

2. Press Alt + F8 to open the Macro dialog.

3. Select Removing_Emoji and click Run.

4. Check multiple cells, including values without trailing emojis, formulas, punctuation, and longer text.

Safety check: A VBA macro may make changes that cannot be reversed with normal Excel Undo. Work on a copy, keep a backup, and only run code you trust.

8. Add a custom Ribbon command

The instructor makes the macro easier to launch by creating a custom Ribbon tab through Excel Options - no custom XML is required for this basic approach.

1. Go to File > Options > Customize Ribbon.

2. Click New Tab; rename it Data Cleaning.

3. Create or rename its custom group SEO Data Cleaner.

4. In Choose commands from, select Macros.

5. Choose Removing_Emoji and click Add to the custom group.

6. Click OK, then use the Ribbon command to run the macro.

Note: Ribbon customizations made through Options may be specific to your Excel installation; they are not guaranteed to accompany the workbook to another computer.

9. Macro Recorder vs Office Scripts

VBA / Macro Recorder	Office Scripts
Uses VBA	Uses TypeScript
Commonly run in desktop Excel	Used in supported Microsoft 365 Excel environments
Can use Alt + F8 macro dialog	Not launched using Alt + F8
Embedded VBA requires .xlsm or .xlsb	Scripts are managed separately

10. Troubleshooting

Issue	What to check
Developer tab missing	File > Options > Customize Ribbon
Macro only changes a few cells	Recorded code uses fixed cell addresses
Macro is missing after saving	Confirm workbook was saved as .xlsm
Wrong worksheet modified	Check whether macro operates on ActiveSheet
Legitimate text removed	Review matching logic and test edge cases
Macro button missing elsewhere	Ribbon customization may be local

11. Lesson revision

Remember: Record a few actions; inspect the VBA; describe the pattern and requirements to AI; review and test the revised code; save as .xlsm; run with Alt + F8; optionally add the macro to a custom Ribbon group.

Knowledge check

1. Where do you enable the Developer tab?
2. What does the Macro Recorder capture?
3. Why might a recorded macro only modify A2:A4?
4. What shortcut opens the Visual Basic Editor?
5. What extension retains VBA macros?
6. What shortcut opens the Macro dialog?
7. Why must you test AI-generated VBA on a copy?
8. How do you add a macro to a custom Ribbon tab?

Answers: 1. File > Options > Customize Ribbon. 2. Supported Excel actions as VBA. 3. It captured specific cell edits. 4. Alt + F11. 5. .xlsm. 6. Alt + F8. 7. Macros can modify data irreversibly or remove valid text. 8. File > Options > Customize Ribbon > New Tab/Group > Choose commands from Macros > Add.

Course progress: Section 31, Video 244 - covered. This PDF summarizes the single lecture supplied for the section.